

Solution of Tutorial on filter bank methods TSIA202b

Roland Badeau

Exercise 1: Capon Estimate of the Parameters of a Single Sine Wave

Assume that the data under study consists of a sinusoidal signal observed in white noise: $\forall t \in \mathbb{Z}$,

$$X_t = \alpha e^{i(2\pi v_0 t + \varphi)} + Z_t$$

where $\alpha \in \mathbb{C}$, $v_0 \in \mathbb{R}$, φ is a random variable uniformly distributed on $[0, 2\pi]$, and $Z_t \sim BB(0, \sigma_Z^2)$ is independent from φ .

1. Prove that X_t is a WSS process and give the mathematical expression of its mean μ_X and its autocovariance function $r_{XX}(k)$.

Firstly, $\mathbb{E}[X_t] = \alpha \int_{\varphi=0}^{2\pi} e^{i(2\pi v_0 t + \varphi)} \frac{d\varphi}{2\pi} + 0 = 0 \forall t \in \mathbb{Z}$ is finite and does not depend on time. Secondly,

$$\text{cov}[X_{t_1}, X_{t_2}] = \mathbb{E} \left[\left(\alpha e^{i(2\pi v_0 t_1 + \varphi)} + Z_{t_1} \right) \left(\overline{\alpha e^{i(2\pi v_0 t_2 + \varphi)} + Z_{t_2}} \right) \right] = |\alpha|^2 e^{i2\pi v_0 (t_1 - t_2)} + \sigma_Z^2 \delta(t_1 - t_2)$$

is finite and only depends on $t_1 - t_2$. Therefore X_t is a WSS process of mean $\mu_X = 0$ and autocovariance function $r_{XX}(k) = |\alpha|^2 e^{i2\pi v_0 k} + \sigma_Z^2 \delta(k)$.

2. Explain the expression line spectrum that is used to name the spectral measure of X_t .

The spectral measure of this WSS process is $|\alpha|^2 (\sum_{n \in \mathbb{Z}} \delta(v - v_0 - n)) + \sigma_Z^2$: its shape is made of one horizontal line, and a countable infinity of vertical lines.

3. Show that the $N \times N$ covariance matrix R_{XX} of $X_1 \dots X_N$ is given by:

$$R_{XX} = |\alpha|^2 e(v_0) e(v_0)^H + \sigma_Z^2 I_N$$

where $e(\xi) = [1, e^{i2\pi\xi} \dots e^{i2\pi\xi(N-1)}]^\top$ and I_N denotes the $N \times N$ identity matrix.

Based on the answer to Question 1, the entry of matrix R_{XX} of indexes (t_1, t_2) is $\text{cov}[X_{t_1}, X_{t_2}] = |\alpha|^2 e^{i2\pi v_0 (t_1 - t_2)} + \sigma_Z^2 \delta(t_1 - t_2)$. In the same way, the entry of matrix $|\alpha|^2 e(v_0) e(v_0)^H + \sigma_Z^2 I_N$ of indexes (t_1, t_2) is $|\alpha|^2 e^{i2\pi v_0 (t_1 - t_2)} + \sigma_Z^2 \delta(t_1 - t_2)$. Therefore $R_{XX} = |\alpha|^2 e(v_0) e(v_0)^H + \sigma_Z^2 I_N$.

4. Check that $\left\{ \frac{e(v_0 + \frac{k}{N})}{\sqrt{N}} \right\}_{k \in \{0 \dots N-1\}}$ forms a unitary basis of eigenvectors of R_{XX} and give the expressions of the corresponding eigenvalues.

We note that:

- $\forall k_1, k_2, \frac{e(v_0 + \frac{k_1}{N})}{\sqrt{N}}^H \frac{e(v_0 + \frac{k_2}{N})}{\sqrt{N}} = \delta(k_1 - k_2)$, therefore $\left\{ \frac{e(v_0 + \frac{k}{N})}{\sqrt{N}} \right\}_{k \in \{0 \dots N-1\}}$ is a unitary basis of $\mathbb{C}^N$;
- $\forall k, R_{XX} \frac{e(v_0 + \frac{k}{N})}{\sqrt{N}} = (|\alpha|^2 e(v_0) e(v_0)^H + \sigma_Z^2 I_N) \frac{e(v_0 + \frac{k}{N})}{\sqrt{N}} = (N|\alpha|^2 \delta(k) + \sigma_Z^2) \frac{e(v_0 + \frac{k}{N})}{\sqrt{N}}$, therefore $\frac{e(v_0 + \frac{k}{N})}{\sqrt{N}}$ is an eigenvector of R_{XX} , of eigenvalue $N|\alpha|^2 \delta(k) + \sigma_Z^2$.

5. Prove that the Capon spectrum $\hat{S}_{\text{CAP},XX}(v) = \frac{N}{e(v)^H R_{XX}^{-1} e(v)}$ peaks at $v = v_0$, and show that the height of the peak is $N|\alpha|^2 + \sigma_Z^2$.

Based on the answer to question 4, we get that the inverse of matrix R_{XX} is

$$R_{XX}^{-1} = \sum_{k=0}^{N-1} \frac{1}{N|\alpha|^2 \delta(k) + \sigma_Z^2} \frac{e(v_0 + \frac{k}{N}) e(v_0 + \frac{k}{N})^H}{N}.$$

Therefore the Capon spectrum is

$$\hat{S}_{\text{CAP},XX}(v) = \frac{N}{\sum_{k=0}^{N-1} \frac{1}{N|\alpha|^2 \delta(k) + \sigma_Z^2} \frac{|e(v)^H e(v_0 + \frac{k}{N})|^2}{N}}.$$

It is maximum when the denominator is minimum, *i.e.* when $v = v_0$. In this case, we get

$$\hat{S}_{\text{CAP},XX}(v_0) = N|\alpha|^2 + \sigma_Z^2.$$